Smaller Diophantine presentations of ZFC consistency

Combining root separation, signed scales, mixed-radix clauses,
and shared evaluation of the nine-variable polynomial

Prepared for Vladimir Reshetnikov's ProveIt repository
OpenAI-assisted mathematical research

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Abstract

We compare the eight wave-3 packages at ProveIt commit `ae3007ca4467` and combine their methods. A new equation in twelve positive unknowns has degree 1081 and a 120-operation circuit, improving our preceding degree-1145, 123-operation equation. A signed-scale extension reduces the moderate-degree unrestricted-integer presentations from 26 to 24 unknowns. Shared evaluation reduces the existing nine-variable polynomial from 20 531 to 1485 operations after fixed coefficient specialization; a sharper majorant gives a different nine-variable equation of lower degree. A mixed-radix compiler reduces the fixed coefficient base to 2^{5^9} and supplies all four actual binary integers. We also give smaller integer low-degree lifts and positive systems of lower degree at unchanged arithmetic cost. These are explicit mathematical constructions with reproducible finite audits, not new end-to-end Lean or Rocq proofs, consistency proofs, or optimality claims.

1 What is minimized, and what the theorem says

There is no single total ordering of "simplest": fewer unknowns, fewer equations, smaller degree, shorter arithmetic circuits, smaller fixed coefficients, and shorter proofs can conflict. We retain a small collection of useful tradeoffs rather than declare one universally best equation. Every operation count charges each binary $+, -, \times$, including multiplication by a fixed numeral. Reused registers and comparisons between already computed expressions are free. Registers are not existential unknowns. All exponentiation in the delivered circuits is unrolled into multiplication; there is no variable exponentiation or hidden divisibility predicate.

Theorem 1 (Meaning of each delivered construction)**.** *For each designated circuit or system P below, with its stated domain $\mathcal{D}$ and its complete fixed coefficient binding, the ordinary mathematical reductions give*

$$\mathrm{ZFC} \vdash \big(\neg\exists \mathbf{x} \in \mathcal{D}\ P(\mathbf{x}) = 0\big) \iff \mathrm{Con}(\mathrm{ZFC}).$$

For a system, $P = 0$ means all its displayed comparisons hold. The assertion is an equivalence, not a proof that either side holds.

The dependency chain is finite syntactic ZFC refutation $\leftrightarrow$ blank halting of the literal K215 machine $\leftrightarrow$ marked periodic tiling $\leftrightarrow$ the inputless native-window/Pell interface $\leftrightarrow$ the equations in their proper domains. The new steps are proved below and in the accompanying detailed notes. The native compiler, Pell-rank lemmas and finitary machine calibration remain cited mathematical dependencies. The source endpoint `KEndpoint215.blank_halts_iff_zfc_refutation` mirrors the last calibration; we do not derive ZFC-provability from Lean's soundness or claim a fresh build. No unconditional two-sided independence statement is made.

2 Comparison with the reviewed sources

The source numbers here are the repository's directory numbers, not machine state counts. The source catalogue was audited at `a477d0fc`; the reviewed HEAD adds proof-simplification proposals but leaves these arithmetic sources unchanged. Their original results are credited, not re-labelled as ours.

Source	Main contribution used	Relation to this synthesis
18-jones9	Concrete parameterized Jones specialization; two-radical finalizer	Same polynomial evaluated much more cheaply; new nonnegative majorants
19-wang943	943 directed-signal tiles; exact-support geometry; materialized $2^{5^{10}}$ tuple	Mixed radix and smaller spatial displacement; same literal language
20-root12	Twelve-positive norm; rational auxiliary-Pell integrality; zero-field mask balancing	Degree/size benchmark; balancing transferred to the 943 language
21-norm12	Economical nested quadratic norms and signed 26-integer form	Separate weights and signed odd scale give new 12-positive and 24-integer forms
22-monic12	Retained-quotient 59-operation system	Exact positive graph lifts preserve 59 operations while lowering degree
23-separating12	Negative-scale exclusion and exact odd-scale completeness	Removes a positivity guard from norm and low-degree presentations
16, 17	Improved auxiliary weights, signed-domain and primitive-element alternatives	Retained as distinct predecessors; not claimed to be invalid

Presentation	Unknowns/domain	Equations	Degree	Operations
Previous own twelve-positive	$12\mathbb{Z}_{>0}$	1	1145	123
New separate weights	$12\mathbb{Z}_{>0}$	1	1081	120
New separate-weight system	$12\mathbb{Z}_{>0}$	3	1072	115
Source 21 integer product	$26\mathbb{Z}$	1	2657	168
New signed-scale baseline	$24\mathbb{Z}$	1	2432	163
New signed-scale split weights	$24\mathbb{Z}$	1	2256	164
New signed-scale ratio weight	$24\mathbb{Z}$	1	1936	168
New baseline comparison system	$24\mathbb{Z}$	3	2416	158
New ratio-weight comparison system	$24\mathbb{Z}$	3	1920	163
Retain positive Y	$17\mathbb{Z}_{>0}$	8	12	59
Retain positive Y, E, c, A_0	$20\mathbb{Z}_{>0}$	11	6	59

The new integer low-degree equations have respectively $51, 42, 36$ unrestricted unknowns at degrees $4, 6, 8$, improving our previous $54, 45, 39$. Their underlying systems have $30, 21, 15$ equations. The existing source-23 27-integer degree-172 product remains useful at lower degree. Likewise P67 remains the least-operation single equation in this comparison; we found no validated single equation below 67 operations or system below 59. The old positive quartic in 32 variables remains useful and is the basis of the nine-variable specialization. No new eight-variable theorem is claimed.

3 A literal common norm evaluator

Fix the four constants (β, K_0, M_C, M_F) from Section 7; M_F is the *shifted* high-mask port. For twelve supplied ports $J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, r$, define

$$\begin{gathered}
q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \quad Y = sq^3, \quad E = XY, \\
k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
\ell = EkY, \quad A = 1 + \ell(\ell + k), \quad G = 1 + \Delta c^2, \\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
N_k = k - hE - R, \qquad N_t = (K_0 + w)C + q - F - z(q-1), \\
I = ic^2, \quad D = 1 + \Delta I^2, \quad Q = (\Delta I)^2, \quad d = Q - 1, \\
v = cT, \quad b = c + RD, \quad e = Q(Dv^2 + b^2) - 1, \quad B_1 = 2Qvb, \\
S = X + ac + Hr.
\end{gathered}$$

Choose weights L, M as specified in the following sections. The norm shorthand

$$N = \prod_{u^2=G,\ \tau^2=A,\ f^2=D} \left[d(S - Lu - HM\tau)^2 - H^2 e + H^2 B_1 f \right] \tag{1}$$

means the product of all eight sign choices, with multiplicity if a radicand vanishes. It is an integer polynomial, not an equation with additional roots. Here is a literal root-free evaluation schedule:

$$\begin{aligned}
a_0 &= H^2 M^2 A, & V_0 &= S^2 + L^2 G, & V_1 &= 2SL, \\
C_0 &= d(V_0 + a_0) - H^2 e, & C_1 &= dV_1, & K_f &= 4d^2 a_0, \\
K_0' &= K_f V_0, & K_1' &= K_f V_1, & Z &= 4H^4 (QDv^2)(Qb^2), \\
U_0 &= C_0^2 + GC_1^2 - K_0' - Z, & U_1 &= 2C_0 C_1 - K_1', & & \\
N_0 &= U_0^2 + GU_1^2 - 4ZK_0', & N_1 &= 2U_0 U_1 - 4ZK_1', & & \\
N &= N_0^2 - GN_1^2. & & & &
\end{aligned}$$

The three finalizations are

$$N_k = N_t = 1, \quad N_0^2 = GN_1^2; \qquad P = N_k N_t (1 + N) - 1; \qquad P_{\text{sos}} = (N_k - 1)^2 + (N_t - 1)^2 + N^2. \tag{2}$$

Counting a system's last equality directly avoids an unnecessary subtraction; this explains the one-gate distinction from residual-zero schedules.

4 Twelve positive unknowns: separate the two weights

All twelve ports are strictly positive. Set

$$W = 2I(T + Raic), \qquad L = 1 + HW, \qquad M = 2W. \tag{3}$$

Using the same large weight for both roots is unnecessary. The main-root weight must be 1 modulo H; the first-root weight only needs sufficient separation. Formula (3) distinguishes these requirements.

At a zero of P, the integer factors in (2) are units. Since N_1 is divisible by 8, $N \equiv N_0^2 \pmod 4$; hence $N = -2$ is impossible, and $N = 0$, $N_k = N_t \in \{-1, 1\}$. Weak transport and the source index bounds give $X \geq 16$, $Y \geq 4096$, $a \geq 64$, $R < c$ and $Q \geq 3$. A vanishing factor in (1) supplies coherent algebraic roots with

$$S = Lu + HM\tau + Hy, \qquad (Q-1)y^2 = Q(vf - b)^2 - 1.$$

Every conjugate, including complex ones, satisfies

$$|y'| < \tfrac{4}{3}(v\sqrt{D}+b) < \tfrac{3}{4}W\sqrt{G}, \qquad \sqrt{G} < \tfrac{9}{8}\sqrt{A}. \tag{4}$$

The first bound follows from $ac < \sqrt{G}$, $\sqrt{D} < (a+2)I$, $a \geq 64$ and $R < c$; the second follows from the same elementary size estimates on X, Y, c. Full rational inequalities are in `norms/TWO_WEIGHT_PROOF.md`. As $HW > 36$,

$$HM\sqrt{A} > \tfrac{16}{9}HW\sqrt{G} > (1+\tfrac{7}{4}HW)\sqrt{G} > L\sqrt{G} + H|y'|.$$

An automorphism flipping an irrational τ would contradict this strict separation. Thus τ is rational and therefore integral. An automorphism flipping irrational u then contradicts $L\sqrt{G} > H|y'|$, so u is integral. Now y is rational and $f = (e - dy^2)/B_1$ is rational and integral; $B_1 > 0$.

Lemma 1 (Special auxiliary-Pell saturation)**.** *If $m \geq 2$, $m, V \in \mathbb{Z}$, and $y \in \mathbb{Q}$, then*

$$m^2V^2 - (m^2-1)y^2 = 1 \quad \Longrightarrow \quad y \in \mathbb{Z}.$$

This is not a claim about arbitrary rational Pell points. Write $m^2 - 1 = D_0 s_0^2$ with D_0 squarefree. Writing $s_0 y = p/q$ in lowest terms, q^2 divides squarefree D_0, so $q = 1$ and $s_0 y \in \mathbb{Z}$. Classify the two integral Pell solutions (m, s_0) and $(|mV|, |s_0 y|)$ by powers of the fundamental unit. The standard divisibility rule for their first coordinates forces the second index to be an odd multiple of the first; the second-coordinate divisibility then gives $s_0 \mid s_0 y$. The supplied source proof also treats the zero and boundary cases. Apply the lemma with $m = |\Delta I|$ to recover integral y.

Reduction modulo H, using $L \equiv 1 \pmod{H}$, recovers $\gamma = (u - X - ac)/H \in \mathbb{Z}$. The restored first, main, strong and auxiliary norms are exactly the source signed-root interface. Its rank/sign argument recovers a genuine blank computation. Conversely choose canonical positive source witnesses and put

$$r = \gamma + Wu + M\tau + y > 0.$$

This annihilates one norm factor and establishes completeness.

For exact degree, put $q_0 = \beta J$, $k_0 = \eta + \zeta$. The leading form is

$$N_{\mathrm{top}} = 2^{32}105^4 i^{48} k_0^{96} w^{80} s^{176} q_0^{656} (q_0 - F)^{16}.$$

It is nonzero for every authentic fixed $\beta > 0$, so $\deg N = 1072$ and $\deg P = 1081$. The tied leading terms must not be silently discarded: the reduced pair calculation gives $C_{0,*} = 19Z_*$ and $K'_{0,*} = 256Z_*^2$, hence the factor 105^4. The JSON schedules have 115, 120, and 123 operations for the system, product, and SOS.

5 Twenty-four unrestricted integer unknowns

Only $J, F, \alpha, z, \eta, \zeta$ retain positivity guards. Replace each by its own copy of

$$\mathrm{Pos}(x, y, t) = 1 + x^2 + y^2 + t(t+1).$$

This is positive and onto $\mathbb{Z}_{>0}$, by the three-square theorem applied to $4n-3$. Set $i = 2i_0+1$, $s = 2s_0+1$, $w = w_0^2$, and let h, T, r be arbitrary integers. There are exactly $6 \cdot 3 + 6 = 24$ unknowns. Positivity of s is *not* assumed during root recovery.

5.1 The least-operation variant and its boundary cases

Set $W = v^2 + D + 2b$, $L = 1 + HW$, $M = L$ in (1). The baseline costs 163 operations at degree 2432; its comparison system costs 158 at maximum degree 2416. Before restoring any roots, weak transport gives $C \geq 0$ and

$$q(q^2-1) < R < q^4 - 2, \quad |Y| \geq q^3, \quad ac > 0, \quad |H| > 6, \quad \Delta > q^4, \quad 1 < b < Q, \quad G > 4.$$

These hold also for negative s and for $w = 0$. Every possible conjugate satisfies $|y'| < W$. If $H < 0$, $L = 1 - |H|W$ is negative, but $|L| > |H|W/2$; thus $|L|\sqrt{G} > |H|W$ still separates the main root.

The $X = 0$ boundary must be excluded *before* using the usual first-root dominance. There $A = 1$, so $\tau = \pm 1$ is already integral; the main-root separation gives integral u and rational y. If $B_1 \neq 0$, recover f and use saturation. If $B_1 = 0$, then $T = v = 0$ and saturation applies directly to $Qb^2 - (Q-1)y^2 = 1$. In either case y and γ are integral. But the main norm at $X = 0$ would give

$$1 = H(2ac\gamma + H\gamma^2 - c^2),$$

impossible for $|H| > 1$. Hence $X \geq q \geq 16$. Now $\sqrt{G} < 4\sqrt{A}$ and $|H| > 6$ imply $|HL|\sqrt{A} > |L|\sqrt{G} + |H|W$. Both roots restore as before. Finally $T = 0$ would imply $(Q-1)(y^2 - b^2) = b^2 - 1$; with $1 < b < Q$, integral y makes this impossible. Thus division by B_1 is justified, not assumed.

5.2 A lower-degree ratio weight

A useful additional tradeoff is

$$W = 2\big[(iI)(1 + 5Rac) + v^2\big], \qquad L = 1 + HW, \qquad M = 2W. \tag{5}$$

Although i, I, a, c may be negative, $iI > 0$ and $ac > 0$, so $W > 0$. The exact cancellation $Dc^2 < GI^2$ gives $\sqrt{D}/\sqrt{G} < |I|/|c|$. Consequently

$$\frac{|y'|}{\sqrt{G}} < 2|iT|c^2 + 2 + 4Rac\,iI < iI + v^2 + 5Rac\,iI = W/2.$$

Here $2|iT| \leq i^2 + T^2$ and $Rac\,iI > 2$ are used; no positive-scale conclusion is used. This restores the main root on the $X = 0$ boundary exactly as above. Once $X > 0$, the uniform estimate

$$\frac{\sqrt{G}}{\sqrt{A}} < \left(1 + \frac{1}{16} + \frac{2}{16 \cdot 32}\right)\left(1 + \frac{1}{32}\right) = \frac{9009}{8192} < \frac{9}{8}$$

and $B_* = |H|W > 8$ imply

$$|L|\sqrt{G} + |H||y'| < \frac{13}{8}\frac{9}{8}B_*\sqrt{A} < 2B_*\sqrt{A}.$$

Thus the first root separates with the smaller multiplier $M = 2W$. The product costs 168 operations and has degree 1936; the system costs 163 with maximum degree 1920. Its leading norm is $89001^4 Z_*^8$, where $Z_* = d_* H_*^2 R_*^2 D_*^2$ has degree 240. The number $89001 = 99 \cdot 899$ comes from tied leading terms, not a generic syntactic-degree estimate. An intermediate split-weight option has degree 2256 and 164 operations.

5.3 Why signed odd scale is enough

After reconstruction, invoke the actual signed-scale theorem from source 23, not a positive-only parent theorem. Its negative-scale argument puts $Y = -Z$, classifies the first and main Pell indices, recovers $p = R$ from the signed auxiliary congruences, and derives $Z > 2^R$. The two possible main-root signs

then give respectively $X \equiv -2^R \pmod{|H|}$ or $X2^R \equiv 1 \pmod{|H|}$, contradicting the strict size bounds. Hence $s > 0$, and source compiler soundness applies.

Completeness needs odd s, not just nonzero s. For canonical witnesses the disjoint packed data and mask give

$$\mathrm{pc}(R) = 3t + 2, \qquad q = 2^t, \qquad v_2(Y) = \mathrm{pc}(R) - 2 = 3t.$$

Thus $s = Y/q^3$ is odd. The same canonical witness simultaneously has odd i and square w; these are not three incompatible witness choices. The new coefficient adapter proves all these promises. For the baseline take $r = \gamma + Wu + L\tau + y$; for (5) take $r = \gamma + Wu + M\tau + y$. No sign restriction on r is needed.

5.4 Low-degree integer lifts

The old low-degree presentations retained a separate s register and three positivity parameters with an equation $s = \mathrm{Pos}$. Replacing that retained register by $2s_0 + 1$ deletes three parameters and one guard equation, without raising any residual degree. This gives 51-integer quartic, 42-integer sextic, and 36-integer octic equations. The saving is three in this retained-register representation, but two in the fully substituted 26-to-24 presentation. The quartic exposes two auxiliary quotients instead of the merged T; the supplied proof restores the signed rank congruences directly, rather than assume their positive quotient proof still applies. Exact coefficient-level substitutions and degree certificates are supplied in `integers/LOW_DEGREE_PROOF.md` and its JSON systems.

6 Nine variables: shorten evaluation, then sharpen the majorant

Source 18 gives a concrete instance of the parameterized `Jones1982.Nine.wset_iff_nine`, with $\nu = 32$ and dummy input $x = 1$, using the normalized positive quartic32. It does not instantiate the separate public theorem that fixes $\nu = 58$. Neither specialization is a new universal small-variable bound. The fixed index constants and polynomial coefficient leaves are finite recipes, not additional unknowns.

6.1 Shared digit-polynomial evaluation

Let $B = b^5$, $T_0 = 5^{32}$, $L_0 = 5T_0$, and $\mathrm{expo}(\alpha) = \sum_{j=0}^{32} \alpha_j 5^j$. The expensive source term is

$$e(B) = \sum_{|\alpha| \le 4} (z_* + c_\alpha) B^{L_0 - \mathrm{expo}(\alpha)}.$$

Define circuit registers $t_j = B^{5^j}$ and $v_j = B^{T_0 - 5^j}$ by $t_{j+1} = t_j^4 t_j$, $v_{32} = 1$, $v_j = t_j^4 v_{j+1}$. Then

$$B^{L_0 - \mathrm{expo}(\alpha)} = t_{32}\, t_{32}^{4-|\alpha|} \prod_j v_j^{\alpha_j}.$$

This identity is polynomial even at $B = 0$, with no division. The dense constant part is a degree-four complete homogeneous polynomial in $(t_{32}, v_0, \ldots, v_{32})$; its 346 sparse corrections can use shared multivariate Horner evaluation. Also

$$\sum_{j=0}^{4L_0-1} B^j = \prod_{j=0}^{32} (1 + t_j + t_j^2 + t_j^3 + t_j^4)(1+q)(1+q^2), \quad q = t_{33},$$

using $1 + t + \cdots + t^4 = 1 + t(1+t)(1+t^2)$. The rest of the source polynomial is unchanged. With independent formal coefficient leaves this reduces 20 531 operations to 1616. Applying the *actual fixed coefficient recipes*, folding their literal integers and merging equal recipes, reduces the count further to 1485. Both variants are supplied, so the extra specialization assumption is explicit and independently checked.

6.2 Sharper natural-domain bounds

In the nine-natural-variable presentation the basic inputs, shifted auxiliary inputs and full digit polynomial are nonnegative. Sign-erasing each subtraction in the packed expression gives $\overline{R} \geq |R|$. Subsequent Jones expressions have nonnegative bounds $\overline{P}, \overline{K}, \overline{A}, \overline{C}, \overline{E}, \overline{G}_p, \overline{H}$, obtained by triangle inequalities. The full formulas appear in `nine/IMPROVEMENT_PROOF.md` and the DAG; nonnegativity of e follows from $z_* + c_\alpha > 0$, not from positivity of its sparse signed corrections. For nonnegative majorants,

$$|(a^2-1)b^2+1| \leq (\bar{a}^2+1)\bar{b}^2+1 \leq ((\bar{a}+1)\bar{b}+1)^2.$$

Use

$$V_0 = (\overline{P}+1)\overline{K}+2,$$
$$V_1 = 2\big[((\overline{A}+1)\overline{C}+1)((\overline{A}+1)\overline{E}+1)((\overline{G}_p+1)\overline{H}+1)\big]+1.$$

These dominate the absolute square roots of the two radicands a_0, a_1 in the source ratio conditions. With its expressions $d > 0, c \geq 0, m$, set

$$\Omega = d(2m-1), \quad \Theta = dn + c - \Omega(c + V_0V_1),$$

$$F_9 = [\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)]^2 - 4\Omega^4 V_0^2 a_0 a_1. \tag{6}$$

The DAG specifies every abbreviation by integer arithmetic. If $F_9 = 0$, the radical combination $r_0 + V_0 r_1 = \Theta/\Omega$ is rational. If nonzero, squaring isolates rational, hence integral, roots. If zero, the strict bound $|r_0| < V_0$ forces $|r_1| < 1$, and both integer radicands vanish. This covers negative radicands, zero roots and dependent square classes. The integer $Z = c + V_0V_1 + r_0 + V_0 r_1$ satisfies $Z \geq c+1$, so $dn + c = d(2m-1)Z$ implies $d \mid c$ and $m > 0$ because $n \geq 0$. Conversely, if both radicands are squares, $d \mid c$ and $m > 0$, they give $n = (2m-1)Z - c/d \geq 1$. Thus the same ratio equivalence applies.

Writing $L_0 = 5^{33}$, the exact new degree is

$$120764L_0 + 368 = 14\,058\,779\,925\,107\,955\,932\,617\,187\,868,$$

versus $161844L_0 + 484$ previously. The diagonal leading coefficient is $9 \cdot 2^{252}$, independent of the fixed compiler tuple; tied terms have leaders 2^{128} and 2^{126} before the final square. The independent-leaf circuit uses 1673 operations; exact fixed-recipe specialization uses 1542. The equivalent positive-domain forms use 1668 and 1537, respectively, after replacing each natural variable by one less than a positive variable and eliminating redundant shifts. Degree is unchanged. These are not unrestricted-integer equations.

7 A fully materialized smaller coefficient tuple

Retain source 19's ordered 943 Wang tiles and occupied ten-bit edge labels. They are reconstructed from all 430 actions of the literal K215 table, including its first action and its original write/move upon entering halt. Only after halt are stationary loops added. The language's marked periodic tiling equivalence and independent powers-of-five padding remain unchanged.

Set $\rho = 2^{125}$, $L_c = 15625$, $d_c = 125L_c = 5^9$, $B_c = \rho^{L_c}$. Let E consist of selector positions $0, \ldots, 942$, forty ten-bit copy positions with offsets $(1023, 1013, 1003, 983)$, dummies $1063, 1064$, and anchors $1095, 1096, 2195, 3296$. Thus $|E| = 989$. Define $w_j = 1024 \cdot 4^j$ for $0 \leq j < 44$. A tile's coefficient is 1 plus the w_j corresponding to its four occupied binary labels, plus $w_{40} + \cdots + w_{43}$. Copy and anchor

coefficients are their respective w_j; dummy coefficients are zero. Put $\mu = 1022 + \sum_{j=0}^{43} w_j$, and define

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e\in E} c_e(\rho^{7631-e} + \rho^{8733-e}),$$

$$D_R = \rho^{42},$$

$$M_F^{\mathrm{nat}} = 4 + \mu(\rho^{7631} + \rho^{8733}) + \sum_{j=0}^{9}(\rho^{1013+j} + \rho^{1065+j}) + \rho^{2195} + \rho^{3296} + \sum_{j=12030}^{12889} \rho^j,$$

$$M_C = B_c - 1 - \sum_{e\in E\setminus\{1\}} \rho^e - 2\rho^{1063}.$$

The four arithmetic constants are

$$\beta = B_c - 1, \quad K_0 = D_C + B_c D_R, \quad M_C, \quad M_F = M_F^{\mathrm{nat}} + B_c - 1. \tag{7}$$

They are fixed by bounded finite recipes and do not depend on waiting for the machine to halt. No coefficient from another tuple is spliced in.

Two distinct carry arguments. The selector count is below 1024, and higher clause digits are multiples of 1024. The low mask 1022 therefore forces count 0 or 1 first. Only then do the 44 base-four payload digits have sums at most 2; their low-bit masks force matching. It would be wrong to assume base-four carry-freedom before selector extraction. Independently, the uniform whole-clause mass satisfies $4\,\mathrm{mass}+8 < 2^{125}$ before any decoding, so there are no radix-ρ carries even for many-selector assignments.

Exact geometry and harmless new tests. The spatial targets are now $1065, \ldots, 1074$ with displacement 42; all unwanted predecessor positions miss E. The two clause bands differ by $1102 \notin E - E$. The two tested anchors have exactly one simultaneous alignment because their difference 1101 occurs only as $3296 - 2195$. The 69 230 exact basis checks verify full coefficient identities, not just interval heuristics. Every aligned unshifted product has exponent at most 12029, so the 860 new mask tests above it are identically zero. They balance population without restricting canonical completeness.

The actual integers satisfy

$$\mathrm{pc}(M_F^{\mathrm{nat}}) = 989, \quad \mathrm{pc}(M_C) = d_c - 989.$$

$$M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8, \quad 2D_C - D_R \equiv 1 \pmod 5.$$

The unit retains the actual-rank dummy tuning. For the exact odd-scale promise, let $q = 2^t$, $J = (q-1)/(B_c - 1)$, and

$$S_0 = (C-1) + qF, \qquad T_0 = (M_C J + 1) + q(M_F^{\mathrm{nat}} J - 1).$$

Native masks give $S_0 \,\&\, T_0 = 0$, both below q^2, and $\mathrm{pc}(T_0) = t + 2$. The packing identity $R = (q^2 - 1 - S_0)q^2 + (S_0 + T_0)$ gives $\mathrm{pc}(R) = 3t + 2$. This is the additional completeness property needed for signed odd s; odd rank alone would not suffice.

Port	Bit length	Minimal little-endian bytes
β	1,953,125	244,141
K_0	1,958,376	244,797
M_C	1,953,125	244,141
M_F (shifted)	1,953,126	244,141
Total		977,220

The binary files and hashes are supplied, not just a compressed recipe. The exponent of B_c is five times smaller than source 19 and twenty-five times smaller than our preceding source 20. Compressed size is not coefficient height. The nine-variable specialization additionally has large fixed index and coefficient recipes; those additional integers are not claimed to be fully materialized. All of them are rebound together to (7) through the delivered fixed-index recipe.

8 Small systems, provenance and verification

The retained source-22 system has sixteen positive variables, seven equations, maximum degree 24 and 59 arithmetic operations. Promoting $Y = sq^3$ to a new positive witness and adding $Y = sq^3$ as a comparison costs no new arithmetic: its old defining gate is still evaluated. It cuts the maximum degree to 12 with seventeen unknowns and eight equations. Retaining Y, E, c, A_0 gives twenty unknowns, eleven equations and degree 6, still at 59 operations. These are exact positive graph lifts, not a hidden redefinition of what counts as an unknown. Their inverse maps are substitution, and the promoted quantities are positive on every old solution. They are useful degree/cost tradeoffs, not new operation minima.

Archive organization. The archive contains this PDF and TeX; exact DAG or residual-polynomial JSON; portable generators and independent checks; detailed universal proof notes; pinned source snapshots and attribution; the complete new coefficient tuple and loader; a whole-tuple instance manifest; and a top-level verifier. The verifier checks sealed hashes and replays calculations in a temporary copy. It must not depend on omitted repository files or a prior working directory. The source catalogue's in-place driver omissions are not defects in the original sealed archives and are avoided here.

What the audits establish. Checks cover circuit syntax and liveness, charged operation ledgers, polynomial identities, signed and degenerate norm fixtures, leading terms, positive shifts, fixed-coefficient aliasing, digit-polynomial support, literal transition and tile equality, exact coefficient geometry, population identities, binary reconstruction, and corruption controls. Independent implementations check key calculations. They do not prove an accepting run, nonhalting, the unbounded compiler theorem, or the metamathematical equivalence merely by passing finite examples. The ordinary proofs, with their explicit dependencies, carry those universal claims. No new formal patch is called kernel-checked, and no compile-time improvement is claimed.

Recommended choice. For fewest variables use the nine-positive-variable construction, accepting its enormous degree and fixed index. For a moderate-degree compact equation use the twelve-positive product. For unrestricted integers choose the 24-variable baseline for arithmetic cost or its ratio-weight variant for degree. For small system arithmetic choose the degree-12, 59-operation positive graph lift. Keep the older P67 and quartic32 when their own metrics matter. These conclusions compare explicit delivered constructions, not all possible Diophantine representations.

Sources and attribution

The reviewed collection is ProveIt, Con(ZFC) in Diophantine form, pinned wave-3 collection. Its eight packages and their inherited references are credited above. The detailed source ledgers give exact paths, pins and hashes. New contributions are the split and ratio weights, signed-scale combinations, shared/homogenized nine-variable evaluation, sign-erased majorants, fixed-recipe circuit folding, mixed-radix clause encoding, smaller displacement, and explicit graph-lift selection. Source licenses and notices are retained. This research was generated with OpenAI assistance for incorporation into ProveIt; it does not attribute human authorship to a model or silently claim authorship of the upstream formalizations.